

Embedded Control Barrier Functions: Concept and Application to Safety-Critical Control Design of High-Relative-Degree Systems

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Abstract—This article proposes a novel safety-critical control (SCC) framework based on embedded control barrier functions (EMB-CBF-SCC) for high-order strict-feedback nonlinear affine control systems. It is aimed at reconciling the potential conflict between predesigned desired trajectory and multiple safety constraints that could have different high relative degrees. Compared with existing CBF-based SCCs, our method can significantly reduce differential order and computational burden. Specifically, we first propose a novel concept of embedded control barrier function (EMB-CBF), which can reduce an arbitrary high-relative-degree safety constraint to relative degree one, and ensure safety of high-relative-degree systems. Further, EMB-CBF-SCC divides the original system into a top-level and a bottom-level subsystem. Then, it embeds between the two subsystems a quadratic program based on EMB-CBF, and introduces command filters to smooth virtual control inputs and obtain differential signals. Coordination performance of safety and stability is analyzed, considering the impact of filter errors. Finally, we present two real safety-critical robotic application scenarios with different safety constraint settings, namely, multiple state constraints for a single-link manipulator numerical model and dynamic obstacle avoidance constraints for a self-developed micro mobile robot experimental platform, respectively. The effectiveness of the proposed framework is demonstrated in both scenarios.

Index Terms—Control barrier function (CBF), high-order system, robot obstacle avoidance, safety-critical control (SCC).

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I. INTRODUCTION

SAFETY is a critical performance index in modern control system design, alongside traditional stability, rapidity, and accuracy criteria. When mobile robots or manipulators perform complex tasks, they are inevitably restricted by external environments or internal joint components. Restrictions to ensure safe operation of these systems are called safety constraints. Several methods have been proposed to deal with safety constraints. Artificial potential field [1] is a real-time navigation algorithm for obstacle avoidance. Prescribed performance control (PPC) [2], [3] and barrier Lyapunov function [4] are control algorithms that transform constrained error signals into unconstrained ones. However, when predesigned desired dynamics conflict with safety constraints, these methods do not allow unsafe system states to occur, resulting in infinite control inputs as systems approach safety boundaries. This severely limits their practical application.

The problem of synthesizing a controller to coordinate safety and stabilization requirements was first presented and coined as safety-critical control (SCC) problem in [5]. Recently, control barrier functions (CBFs) have become popular [6] in applications, such as autonomous ground vehicles [7] and autonomous surface vehicles [8]. CBFs allow systems to be in an unsafe situation in initial and intermediate states. However, they are only suitable for safety constraints with relative degree one. For higher relative degrees, the control input becomes invalid [9], [10].

Practical robotic systems are often high-relative-degree systems, i.e., system models have high-order forms (second-order or higher), and their safety constraints have also high relative degrees (2 or higher). For these systems, the notion of high-order control barrier function (HOCBF) was proposed in [10], by constructing auxiliary functions. The HOCBF concept was further generalized in [11], where a singularity-free control framework was developed. HOCBF was extended to discrete-time systems in [12], and its application to compliantly actuated robots was reported in [13]. On the other hand, an alternative safe backstepping (SBP) approach was proposed in [14], by recursively designing safe virtual control inputs with explicit expressions. Both HOCBF and SBP can meet SCC objectives in high-relative-degree systems, but they require high-order derivatives of safety constraints. Safe velocity tracking (SVT) was proposed in [15]

for second-order robotic systems. It does not rely on high-fidelity dynamical models, but cannot be directly extended to general higher relative-degree systems. To sum up, achieving SCC for high-relative-degree systems with low computational burden and high generality is important and challenging [16].

This article introduces a novel concept of embedded control barrier function (EMB-CBF, as defined by Definition 5 in Section III), aimed at ensuring safety of high-relative-degree systems and reducing the relative degree of safety constraints. In addition, this article proposes a novel safety-critical control framework based on EMB-CBF (EMB-CBF-SCC) to achieve coordination of safety and stability (as per Definition 6 in Section IV) for high-order strict-feedback nonlinear affine control systems with multiple safety constraints that could have different high relative degrees. Specifically, the system is divided into two subsystems and a quadratic program based on EMB-CBF (EMB-CBF-QP) is embedded between them. Both numerical simulations and practical experiments demonstrate the effectiveness of EMB-CBF-SCC.

The main contributions of this article are as follows.

- 1) EMB-CBF reduces single arbitrary high-relative-degree safety constraints to relative degree one, which allows more flexible design and simpler computation than existing methods.
- 2) EMB-CBF-SCC significantly reduces differential order and computational burden compared with existing CBF-based methods, and for the first time (to the best of the authors' knowledge) addresses multiple safety constraints with different high relative degrees. Compared with HOCBF [10], [11] and SBP [14], the location change of quadratic program makes our proposed framework require only the lower order derivative of safety constraints instead of higher order ones. Compared with SVT [15], the proposed method can be applied to second-order or higher systems, achieving better performance through control decoupling of subsystems.
- 3) Using command filters for EMB-CBF-SCC, to avoid differential explosion and smooth the output signal of EMB-CBF-QP, alleviates the issue that the solution of EMB-CBF-QP cannot be strictly guaranteed to be differentiable.

The rest of the article is organized as follows. Section II describes the preliminaries. Section III introduces the notion of the proposed EMB-CBF. Section IV presents the proposed EMB-CBF-SCC framework. Section V presents numerical simulations and real experiments for two robotic application scenarios to validate the work. Finally, Section VI concludes this article and points to future work.

II. PRELIMINARIES

Notations: $\|\cdot\|$ denotes the Euclidean norm of a vector or matrix, $\lambda_{\min}(\cdot)$ is the minimum eigenvalue of a matrix, L_f is the Lie derivative along f , L_f^m is the m times Lie derivative along f , $\mathcal{K}^e$ is the extended class $\mathcal{K}$ function, and $\mathcal{K}_\infty^e$ is the extended class $\mathcal{K}_\infty$ function.

Let us consider an affine control systems of the form

$$\dot{x} = f(x) + g(x)u \quad (1)$$

where $x \in \mathcal{X} \subset \mathbb{R}^n$ and $u \in \mathcal{U} \subset \mathbb{R}^p$ represent system state and control input, respectively, $f : \mathcal{X} \rightarrow \mathbb{R}^n$ and $g : \mathcal{X} \rightarrow \mathbb{R}^{n \times p}$ are locally Lipschitz, and system solutions are forward complete. Let $h(x) \geq 0$ be the safety constraint. We define

$$\begin{aligned} \mathcal{C} &= \{x \in \mathcal{X} : h(x) \geq 0\} \\ \partial\mathcal{C} &= \{x \in \mathcal{X} : h(x) = 0\} \end{aligned} \quad (2)$$

where $\partial\mathcal{C}$ is the boundary of superlevel set $\mathcal{C}$.

Definition 1: Set $\mathcal{C}$ is forward invariant for system (1) if its solutions starting at any $x(t_0) \in \mathcal{C}$ satisfy $x(t) \in \mathcal{C} \forall t \geq t_0$. System (1) is safe with respect to safe set $\mathcal{C}$.

Definition 2: Set $\mathcal{C}$ is the superlevel set of a continuously differentiable function $h : \mathcal{X} \rightarrow \mathbb{R}$ if there exists $\gamma \in \mathcal{K}^e$ such that for $\forall x \in \mathcal{X}$

$$\sup_{u \in \mathcal{U}} [L_f h(x) + L_g h(x)u + \gamma(h(x))] \geq 0 \quad (3)$$

then h is a CBF for system (1).

Definition 3: Let origin $x^* = 0 \in \mathcal{X}$ be the equilibrium point. Given continuously differentiable function $V : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$, if there exists $\alpha_1, \alpha_2, \alpha_3 \in \mathcal{K}$ such that for $\forall x \in \mathcal{X}$,

$$\alpha_1(\|x\|) \leq V(x) \leq \alpha_2(\|x\|) \quad (4)$$

$$\inf_{u \in \mathcal{U}} [L_f V(x) + L_g V(x)u + \alpha_3(V(x))] \leq 0 \quad (5)$$

then V is a control Lyapunov function (CLF) for system (1).

According to the definition of relative degree in [9], for r th-order differentiable function $h : \mathcal{X} \rightarrow \mathbb{R}$ with high relative degree $r \geq 2$, CBF is invalid. In this case, a sequence of auxiliary functions $\psi_i : \mathcal{X} \rightarrow \mathbb{R}$ can be constructed as

$$\begin{aligned} \psi_0(x) &= h(x) \\ \psi_i(x) &= \dot{\psi}_{i-1}(x) + \gamma_i(\psi_{i-1}(x)), i = 1, \dots, r \end{aligned} \quad (6)$$

where $(r-i)$ th-order differentiable function $\gamma_i \in \mathcal{K}^e$. The superlevel sets of these auxiliary functions are

$$\mathcal{C}_i = \{x \in \mathcal{X} : \psi_{i-1}(x) \geq 0\}, i = 1, \dots, r. \quad (7)$$

Definition 4: Given $\mathcal{C}_i$ from (7) and $\psi_i : \mathcal{X} \rightarrow \mathbb{R}$ from (6), if there exists $(r-i)$ th-order differentiable function $\gamma_i \in \mathcal{K}^e$ ($i = 1, \dots, r$), such that for $\forall x \in \mathcal{X}$

$$\sup_{u \in \mathcal{U}} [L_f^r h(x) + L_g L_f^{r-1} h(x)u + O(h(x)) + \gamma_r(\psi_{r-1}(x))] \geq 0 \quad (8)$$

where $O(h(x)) = \sum_{i=1}^{r-1} L_f^i (\gamma_{r-i} \circ \psi_{r-i-1})(x)$ represents the residual Lie derivative, then h is an HOCBF of relative degree r for system (1).

HOCBF ensures the forward invariance of set $\mathcal{C}_1 \cap \mathcal{C}_2 \cap \dots \cap \mathcal{C}_r$ [10]. It requires the constraint to be r th-order continuously differentiable, and involves complicated differential expressions up to r th-order, which cause differential explosion.

III. EMBEDDED CBFs

Let us assume that $x = [x_T^\top, x_B^\top]^\top$ and system (1) can be divided into two parts

$$\begin{cases} \dot{x}_T = f_T(x_T) + g_T(x_T)x_B \\ \dot{x}_B = f_B(x_T, x_B) + g_B(x_T, x_B)u \end{cases} \quad (9)$$

where $x_T \in \mathcal{X}_T$, $x_B \in \mathcal{X}_B$, and $u \in \mathcal{U}$. The dynamics of x_T and x_B are called top-level subsystem (TLS) and bottom-level subsystem (BLS), respectively. The safety constraint can be rewritten as $h(x_T) \geq 0$, and satisfies $L_{g_T}h(x_T) \neq 0$ for $\forall x_T \in \mathcal{X}_T$. Thus, the original superlevel set is equivalent to

$$\mathcal{C} = \{x_T \in \mathcal{X}_T : h(x_T) \geq 0\}. \quad (10)$$

Inspired by backstepping method, we define a continuously differentiable intermediate virtual control input v_T to replace state x_B as the direct control input of TLS. The tracking error of BLS is $z_B = x_B - v_T$, so system (9) can be converted to

$$\begin{cases} \dot{x}_T = f_T(x_T) + g_T(x_T)(v_T + z_B) \\ \dot{z}_B = f_B(x_T, z_B) + g_B(x_T, z_B)u - \dot{v}_T \end{cases} \quad (11)$$

where the dynamics of z_B is called bottom-level error subsystem (BLES) with equilibrium point $z_B^* = 0$. We define a positive definite continuously differentiable function $V_B(z_B) : \mathbb{R}^{|\mathcal{X}_B|} \rightarrow \mathbb{R}_{\geq 0}$ as a CLF of BLES. By subtracting the CLF of BLES from the original safety constraint, we construct the modified safety constraint $H(x_T, z_B)$ as follows:

$$H(x_T, z_B) = h(x_T) - \mu V_B(z_B) \quad (12)$$

where $\mu > 0$ is an adjustable parameter. The superlevel set corresponding to the modified safety constraint is

$$\mathcal{C}_E = \{x_T \in \mathcal{X}_T, z_B \in \mathbb{R}^{|\mathcal{X}_B|} : H(x_T, z_B) \geq 0\}. \quad (13)$$

Obviously, direct control input u appears explicitly in the differential expression of $H(x_T, z_B)$, which means that the relative degree of single safety constraint is reduced from arbitrary high to one. Thus, EMB-CBF is defined as follows.

Definition 5: Given the TLS and BLS from (9), $\mathcal{C}$ from (10), and $\mathcal{C}_E$ from (13) are the superlevel sets of continuously differentiable functions $h : \mathcal{X}_T \rightarrow \mathbb{R}$ and $H : \mathcal{X}_T \times \mathbb{R}^{|\mathcal{X}_B|} \rightarrow \mathbb{R}$ from (12), respectively. If there exists $\gamma \in \mathcal{K}_\infty^e$ and continuously differentiable intermediate virtual control input $v_T \in \mathcal{X}_B$, such that for $\forall x_T \in \mathcal{X}_T$ and $\forall x_B \in \mathcal{X}_B$

$$\sup_{v_T \in \mathcal{X}_B, u \in \mathcal{U}} [\dot{H}(x_T, z_B, v_T, \dot{v}_T, u) + \gamma(H(x_T, z_B))] \geq 0 \quad (14)$$

where

$$\begin{aligned} \dot{H}(x_T, z_B, v_T, \dot{v}_T, u) = & L_{f_T}h(x_T) + L_{g_T}h(x_T)(v_T + z_B) \\ & - \mu L_{f_B}V_B(z_B) - \mu L_{g_B}V_B(z_B)u + \mu L_{\dot{v}_T}V_B(z_B) \end{aligned} \quad (15)$$

then H is an EMB-CBF for system (9).

Theorem 1: Given the TLS and BLS from (9), $\mathcal{C}$ from (10), and $\mathcal{C}_E$ from (13) are the superlevel sets of continuously differentiable functions $h : \mathcal{X}_T \rightarrow \mathbb{R}$ and $H : \mathcal{X}_T \times \mathbb{R}^{|\mathcal{X}_B|} \rightarrow \mathbb{R}$ from (12), respectively. If H is an EMB-CBF and $\frac{\partial h(x_T)}{\partial x_T} \neq 0$ for $\forall x_T \in \partial \mathcal{C}$, then any continuously differentiable intermediate

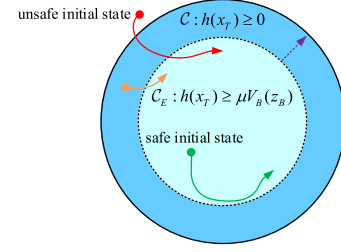


Fig. 1. Intuitive description of EMB-CBF.

virtual controller v_T and Lipschitz continuous controller u satisfying (14) can ensure the forward invariance of set $\mathcal{C}_E$ and the safety of system (9).

Proof: Condition (14) means that continuously differentiable intermediate virtual control input $v_T \in \mathcal{X}_B$ and direct control input $u \in \mathcal{U}$ satisfy $\dot{H}(x_T, z_B, v_T, \dot{v}_T, u) + \gamma(H(x_T, z_B)) \geq 0$. Further, $H(x_T, z_B)$ can be proved to satisfy boundary condition: $[\frac{\partial H(x_T, z_B)}{\partial x_T}, \frac{\partial H(x_T, z_B)}{\partial z_B}] \neq 0$ for $\forall (x_T, z_B) \in \partial \mathcal{C}_E$ by contradiction. Otherwise, we assume that $\exists (x'_T, z'_B) \in \partial \mathcal{C}_E$ such that $[\frac{\partial H(x'_T, z'_B)}{\partial x'_T}, \frac{\partial H(x'_T, z'_B)}{\partial z'_B}] = 0$, then from (12) $\frac{\partial h(x'_T)}{\partial x'_T} = \frac{\partial H(x'_T, z'_B)}{\partial x'_T} = 0$ and $\frac{\partial V_B(z'_B)}{\partial z'_B} = -\frac{1}{\mu} \frac{\partial H(x'_T, z'_B)}{\partial z'_B} = 0$. Since V_B is a CLF of BLES, according to Definition 3, there exists $\alpha_1, \alpha_2, \alpha_3 \in \mathcal{K}$ such that $L_{f_B}V_B(z'_B) + L_{g_B}V_B(z'_B)u - L_{\dot{v}_T}V_B(z'_B) + \alpha_3(V_B(z'_B)) = \alpha_3(V_B(z'_B)) \leq 0$. Meanwhile, $0 \leq \alpha_1(\|z'_B\|) \leq V_B(z'_B) \leq \alpha_2(\|z'_B\|)$, $\alpha_3(V_B(z'_B)) \geq 0$ due to the property of class $\mathcal{K}$ function, thus $\alpha_3(V_B(z'_B)) = 0$, $V_B(z'_B) = 0$, and $z'_B = 0$. Since $(x'_T, z'_B) \in \partial \mathcal{C}_E$, one has $h(x'_T) = H(x'_T, z'_B) = 0$, so $x'_T \in \partial \mathcal{C}$, and $\frac{\partial h(x'_T)}{\partial x'_T} \neq 0$ due to the boundary condition of h , which contradicts $\frac{\partial h(x'_T)}{\partial x'_T} = 0$.

Thus, EMB-CBF $H(x_T, z_B)$ is also a CBF for entire system (9), according to Definition 2. It ensures the forward invariance of set $\mathcal{C}_E$ according to [6, Thm. 2], i.e., if initial state $(x_T(t_0), z_B(t_0)) \in \mathcal{C}_E$, then $x_T(t) \in \mathcal{C}$ for $\forall t \geq t_0$ because $h(x_T) \geq H(x_T, z_B) \geq 0$, thus safety of system (9) is ensured. The proof is complete.

Fig. 1 shows an intuitive description of EMB-CBF. EMB-CBF limits the safety constraint from original safety set $\mathcal{C}$ to a tighter modified safety set $\mathcal{C}_E$, but $\mathcal{C}_E \rightarrow \mathcal{C}$ as $z_B \rightarrow 0$ (convergence of BLES). The unsafe initial state starting from external of $\mathcal{C}_E$ will first reach $\mathcal{C}$, and then $\mathcal{C}_E$; the safe initial state starting from interior of $\mathcal{C}_E$ will always stay in $\mathcal{C}_E$. Parameter μ regulates the range of $\mathcal{C}_E$. There are two types of control inputs in (15): intermediate virtual control input v_T and direct control input u . Thus, we can adopt expression-based controller (EBC) or optimization-based controller (OBC) for them independently. EBC means that the subsystem controller has an explicit expression, whereas OBC means that the subsystem obtains the control input by solving an optimization problem. Since $H(x_T, 0) = h(x_T)$ has relative degree one and v_T has affine form with respect to TLS, it is feasible that v_T adopts OBC and u adopts EBC. This is the fundamental difference between the EMB-CBF-SCC proposed in Section IV and other CBF-based methods.

From the comparisons shown in Table I, CBF-QP is applicable to safety constraints with relative degree 1; the original studies of

TABLE I
COMPARISON OF EMB-CBF-SCC WITH CBF-BASED METHODS

Method	TLS	BLS	RD	DO
CBF-QP [5]	—	OBC	1	1
HOCBF-QP [10]	—	OBC	High	High
SBP [14]	EBC	OBC	High	High
SVT [15]	OBC	EBC	2	1
ML-SCC [17]	OBC	OBC	2	1
EMB-CBF-SCC	OBC	EBC	High	1

* RD: safety constraint relative degree applied by the method.

DO: safety constraint differential order required by the method.

SVT and multilayered safety-critical control (ML-SCC) focus on second-order robotic systems only, and are applicable to safety constraints with relative degree two; SBP is applicable to high-relative-degree systems where the relative degrees of safety constraints equal the system order; only HOCBF-QP and EMB-CBF-SCC are suitable for high-relative-degree systems where safety constraints have arbitrarily high relative degrees. In terms of computational burden, SBP relies on recursively designing safe intermediate virtual control inputs with explicit expression and HOCBF-QP does not involve subsystem division operation. Thus, they both require high-order derivatives of safety constraints, which can easily cause “calculation explosion.” OBC is necessary in at least one of the two subsystem controllers in order to coordinate system stability and safety. Both SVT and ML-SCC adopt OBC in their TLSs, so they only require first-order derivatives of safety constraints, whereas two OBCs in ML-SCC bring extra computational burden. Essentially, EMB-CBF-SCC generalizes and extends SVT’s framework, by using OBC in TLS and EBC in BLS. It is suitable for high-relative-degree systems, and only requires first-order derivatives of safety constraints. This results in more flexible design and simpler computation. OBC seems to be embedded between the two subsystems, and hence the nomenclature “EMB-CBF.”

IV. SCC DESIGN BASED ON EMB-CBF

A. Coordination of Safety and Stability

In SCC design, the system must always meet the safety constraints (high priority), whereas the stability requirement is that the system tracks predesigned desired trajectories (low priority). These requirements may be contradictory. To properly handle this potential contradiction, the following definition allows us to separately prove the safety and stability of a safety-critical controller.

Definition 6: A control system is said to achieve coordination of safety and stability if the system with safety constraints is safe and the system without safety constraints is stable.

B. Problem Formulation

Let us consider a class of high-order strict-feedback nonlinear affine control systems as follows:

$$\begin{cases} \dot{x}_i = f_i(\bar{x}_i) + g_i(\bar{x}_i)x_{i+1}, i = 1, \dots, n-1 \\ \dot{x}_n = f_n(\bar{x}_n) + g_n(\bar{x}_n)u \\ y = x_1 \end{cases} \quad (16)$$

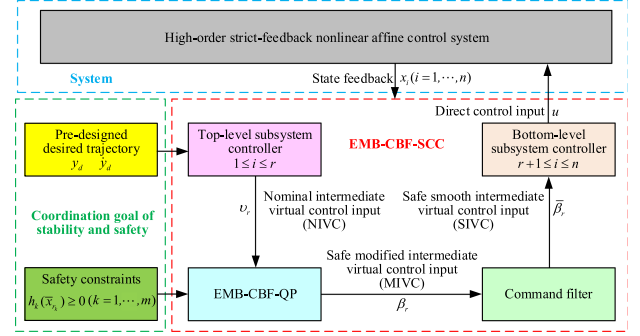


Fig. 2. Structure of EMB-CBF-SCC.

where $x_i \in \mathcal{X}_i \subset \mathbb{R}^{p_i}$ ($i = 1, \dots, n$) is the state vector of the i th subsystem, $\bar{x}_i = [x_1^\top, \dots, x_i^\top]^\top \in \mathcal{W}_i \triangleq \mathcal{X}_1 \times \dots \times \mathcal{X}_i$ is the compound state vector, and $u \in \mathcal{U} \subset \mathbb{R}^{p_{n+1}}$ and $y \in \mathcal{X}_1$ are the control input and output vectors, respectively. Functions $f_i : \mathcal{W}_i \rightarrow \mathbb{R}^{p_i}$ and $g_i : \mathcal{W}_i \rightarrow \mathbb{R}^{p_i \times p_{i+1}}$ are smooth in their respective domains. Solutions $x_i(t)$ of system (16), starting at $x_i(t_0) \in \mathcal{X}_i$, $t \geq t_0$, are forward complete.

Assumption 1: g_i has full row rank, thus its Moore–Penrose (right) pseudo-inverse is $g_i^\dagger = g_i^\top (g_i g_i^\top)^{-1}$. All elements of f_i , g_i , $g_i^\dagger$, $\dot{f}_i$, $\dot{g}_i$, and $\dot{g}_i^\dagger$ are bounded.

Assumption 2: Predesigned desired trajectory $y_d \in \mathcal{X}_1$ may conflict with constraints. y_d and $\dot{y}_d$ are continuous, bounded, and available. $\ddot{y}_d$ is also continuous and bounded.

Assumption 3: The m safety constraints of the system can be expressed as $h_k(\bar{x}_{r_k}) \geq 0$ ($k = 1, \dots, m$) with relative degree $n - r_k + 1$, where $h_k : \mathcal{W}_{r_k} \rightarrow \mathbb{R}$ is a sufficient times differentiable function and $1 \leq r_k \leq n - 1$.

SCC objective is to design u to achieve coordination of safety and stability, i.e., 1) $h_k(\bar{x}_{r_k}(t)) \geq 0$ ($k = 1, \dots, m$) as $t \geq t_0$. 2) $y_1(t) \rightarrow y_d(t)$ as $t \rightarrow \infty$ for the system without safety constraints.

C. SCC Design

The structure of EMB-CBF-SCC is shown in Fig. 2. Let $r = \max_{1 \leq k \leq m} r_k$. Since all safety constraints are independent of the last $n - r$ system states, we divide the first r subsystems into TLS, and the last $n - r$ ones into BLS. Thus, the k th safety constraint has only relative degree $r - r_k + 1$ with respect to TLS. Designing EMB-CBF-SCC has three steps. 1) For stability of top-level error subsystem (TLES) without safety constraints, recursively design virtual control inputs $v_1, \dots, v_{r-1}$ and nominal intermediate virtual control (NIVC) input v_r . 2) Use EMB-CBF-QP to transform NIVC v_r into safe modified intermediate virtual control (MIVC) input β_r , to guarantee safety of TLS with safety constraints. MIVC is smoothed by the command filter to obtain safe smooth intermediate virtual control input (SIVC) $\bar{\beta}_r$ and its differential signal $\dot{\bar{\beta}}_r$. 3) For BLES convergence, recursively design virtual control inputs $v_{r+1}, \dots, v_{n-1}$ and direct control input u .

Inspired by command-filtered backstepping [18], we introduce a series of first-order command filters to avoid the repeated

differentiation of virtual control inputs

$$\dot{\bar{v}}_i = \tau_i(v_i - \bar{v}_i), i = 1, \dots, r-1, r+1, \dots, n-1 \quad (17)$$

where τ_i is a positive definite diagonal constant matrix, v_i and $\bar{v}_i$ are the input and output signals of the command filter, respectively, the initial condition is $\bar{v}_i(t_0) = v_i(t_0)$, $\dot{\bar{v}}_i$ is the differential signal, and $\tilde{v}_i = \bar{v}_i - v_i$ is the filter error. Similarly, we define $\tilde{\beta}_r = \bar{\beta}_r - \beta_r$.

Performing the following coordinate transformation:

$$\begin{aligned} z_1 &= y - y_d \\ z_i &= x_i - \bar{v}_{i-1}, i = 2, \dots, r, r+2, \dots, n \\ z_{r+1} &= x_{r+1} - \bar{\beta}_r \end{aligned} \quad (18)$$

then TLES can be represented as follows:

$$\begin{aligned} \dot{z}_1 &= f_1 + g_1(v_1 + \tilde{v}_1 + z_2) - \dot{y}_d \\ \dot{z}_i &= f_i + g_i(v_i + \tilde{v}_i + z_{i+1}) - \dot{\bar{v}}_{i-1}, i = 2, \dots, r-1 \\ \dot{z}_r &= f_r + g_r(\beta_r + \tilde{\beta}_r + z_{r+1}) - \dot{\bar{v}}_{r-1} \end{aligned} \quad (19)$$

and BLES can be represented as follows:

$$\begin{aligned} \dot{z}_{r+1} &= f_{r+1} + g_{r+1}(v_{r+1} + \tilde{v}_{r+1} + z_{r+2}) - \dot{\tilde{\beta}}_r \\ \dot{z}_i &= f_i + g_i(v_i + \tilde{v}_i + z_{i+1}) - \dot{\bar{v}}_{i-1}, i = r+2, \dots, n-1 \\ \dot{z}_n &= f_n + g_n u - \dot{\bar{v}}_{n-1}. \end{aligned} \quad (20)$$

To avoid introducing multiple QPs to handle multiple safety constraints with different relative degrees, inspired by HOCBF, we unify safety constraints into a single QP by constructing the following auxiliary functions for the k th safety constraint:

$$\begin{aligned} \psi_{0,k}(\bar{x}_{r_k}) &= h_k(\bar{x}_{r_k}) \\ \psi_{j,k}(\bar{x}_{r_k+j}) &= \dot{\psi}_{j-1,k}(\bar{x}_{r_k+j}) + \gamma_{j,k}(\psi_{j-1,k}(\bar{x}_{r_k+j-1})) \\ j &= 1, \dots, r-r_k \end{aligned} \quad (21)$$

where $(r-r_k+1-j)$ th-order is the differentiable function $\gamma_{j,k} \in \mathcal{K}^e$. The superlevel sets of these auxiliary functions are

$$\begin{aligned} \mathcal{C}_{j,k} &= \left\{ \bar{x}_r \in \mathcal{W}_r, z_B \in \mathbb{R}^{|\mathcal{X}_B|} : \psi_{j-1,k}(\bar{x}_{r_k+j-1}) \geq 0 \right\} \\ j &= 1, \dots, r-r_k+1 \end{aligned} \quad (22)$$

where $\mathcal{X}_B \triangleq \mathcal{X}_{r+1} \times \dots \times \mathcal{X}_n$. The continuously differentiable SIVC $\bar{\beta}_r$ explicitly appears and has affine form in the time derivative of $\psi_{r-r_k,k}(\bar{x}_r)$

$$\begin{aligned} &\dot{\psi}_{r-r_k,k}(\bar{x}_r, \bar{\beta}_r, z_{r+1}) \\ &= \sum_{i=1}^{r-1} \frac{\partial \psi_{r-r_k,k}}{\partial x_i} (f_i + g_i x_{i+1}) + \frac{\partial \psi_{r-r_k,k}}{\partial x_r} [f_r + g_r(\bar{\beta}_r + z_{r+1})] \\ &= \dot{\psi}_{r-r_k,k}(\bar{x}_r, \beta_r, 0) + \frac{\partial \psi_{r-r_k,k}}{\partial x_r} (\tilde{\beta}_r + z_{r+1}). \end{aligned} \quad (23)$$

In addition, we select the CLF of TLES as $V_T(z_T) = 0.5z_T^\top z_T$, and that of BLES as $V_B(z_B) = 0.5z_B^\top z_B$, where $z_T = [z_1^\top, \dots, z_r^\top]^\top$ and $z_B = [z_{r+1}^\top, \dots, z_n^\top]^\top$ are the compound state

vectors of TLES and BLES, respectively. Thus, according to (12), one can construct the EMB-CBF

$$H_k(\bar{x}_r, z_B) = \psi_{r-r_k,k}(\bar{x}_r) - \mu V_B(z_B) \quad (24)$$

where $\mu > 0$ is a designed parameter. The superlevel set corresponding to the EMB-CBF (24) is

$$\mathcal{C}_{E,k} = \left\{ \bar{x}_r \in \mathcal{W}_r, z_B \in \mathbb{R}^{|\mathcal{X}_B|} : H_k(\bar{x}_r, z_B) \geq 0 \right\}. \quad (25)$$

The three steps in SCC design are described in detail below.

Step 1: For $i = 1$, the virtual control input is

$$v_1 = -g_1^\dagger(\kappa_1 z_1 + f_1 - \dot{y}_d) \quad (26)$$

where κ_1 is a designed positive definite matrix. $\bar{v}_1$ and $\dot{\bar{v}}_1$ are obtained by command filter (17).

For $2 \leq i \leq r-1$, the virtual control input is

$$v_i = -g_i^\dagger(\kappa_i z_i + f_i - \dot{\bar{v}}_{i-1} + g_{i-1}^\top z_{i-1}) \quad (27)$$

where κ_i is a designed positive definite matrix. $\bar{v}_i$ and $\dot{\bar{v}}_i$ are obtained by command filter (17).

For $i = r$, NIVC is

$$v_r = -g_r^\dagger(\kappa_r z_r + f_r - \dot{\bar{v}}_{r-1} + g_{r-1}^\top z_{r-1}) - \delta g_r^\top z_r \quad (28)$$

where κ_r is a designed positive definite matrix and $\delta > 0$ is a designed parameter.

Step 2: Synthesizing an OBC to obtain MIVC, i.e., solving the following EMB-CBF-QP:

$$\begin{aligned} \beta_r(\bar{x}_r) &= \arg \min_{\beta_r \in \mathcal{X}_r} \|\beta_r - v_r\| \\ \text{s.t. } &\dot{\psi}_{r-r_k,k}(\bar{x}_r, \beta_r, 0) + \gamma_{r-r_k+1,k}(\psi_{r-r_k,k}(\bar{x}_r)) \geq \\ &\frac{2\delta}{\mu} \frac{\partial \psi_{r-r_k,k}}{\partial x_r} g_r g_r^\top \left(\frac{\partial \psi_{r-r_k,k}}{\partial x_r} \right)^\top, k = 1, \dots, m \end{aligned} \quad (29)$$

where $\gamma_{r-r_k+1,k} \in \mathcal{K}_\infty^e$ is a globally Lipschitz function with Lipschitz constant $l_{\gamma,k} > 0$.

MIVC cannot guarantee the differentiable property required by Definition 5. To address this issue, we use a first-order command filter to obtain a smooth approximate signal:

$$\dot{\tilde{\beta}}_r = \tau_r(\beta_r - \bar{\beta}_r) \quad (30)$$

where τ_r is a positive definite diagonal constant matrix. $\bar{\beta}_r(t_0) = \beta_r(t_0)$ is the initial condition, output signal $\bar{\beta}_r$ is the SIVC, and $\dot{\tilde{\beta}}_r$ is the differential signal.

Step 3: For $i = r+1$, the virtual control input is

$$v_{r+1} = -g_{r+1}^\dagger \left(\kappa_{r+1} z_{r+1} + f_{r+1} - \dot{\tilde{\beta}}_r + \frac{1}{4\delta} z_{r+1} \right) \quad (31)$$

where κ_{r+1} is a designed positive definite matrix. $\bar{v}_{r+1}$ and $\dot{\bar{v}}_{r+1}$ are obtained by command filter (17).

For $r+2 \leq i \leq n-1$, the virtual control input is

$$v_i = -g_i^\dagger(\kappa_i z_i + f_i - \dot{\bar{v}}_{i-1} + g_{i-1}^\top z_{i-1}) \quad (32)$$

where κ_i is a designed positive definite matrix. $\bar{v}_i$ and $\dot{\bar{v}}_i$ are obtained by command filter (17).

Finally, for $i = n$, the direct control input is

$$u = -g_n^\dagger(\kappa_n z_n + f_n - \dot{\bar{v}}_{n-1} + g_{n-1}^\top z_{n-1}) \quad (33)$$

where κ_n is a designed positive definite matrix.

Remark 1: Compared with HOCBF, EMB-CBF-SCC only requires the $(r - r_k + 1)$ th-order derivative of the k th safety constraint instead of the $(n - r_k + 1)$ th-order one. Thus, it reduces $(n - r)$ th-order differential operation and calculation.

Remark 2: Compared with classical backstepping, we introduce additional terms $-\delta g_r^\top z_r$ in v_r (28) and $-\frac{1}{4\delta} g_{r+1}^\top z_{r+1}$ in v_{r+1} (31) to realize the control decoupling of subsystems. EMB-CBF-QP (29) introduces two additional parameters, namely, δ and μ . As μ decreases or δ increases, system safety improves but the feasible region of EMB-CBF-QP is reduced, the arrival speed of safe set increases when system starts from unsafe initial states.

Remark 3: In this article, system model (16) is fully known. For more general systems with unknown terms, uncertainties, and disturbances, although the proposed EMB-CBF-SCC exhibits some robustness, a further improvement is to combine it with estimation algorithms, such as RBF neural networks or extended state observers.

D. Safety and Stability Analysis

Lemma 1 (Infer from [18]): For first-order command filter

$$\dot{\bar{v}}(t) = \tau[v(t) - \bar{v}(t)], t \geq t_0 \quad (34)$$

where $v(t)$ is a continuously differentiable reference signal vector, τ is a positive definite matrix, and $\bar{v}(t_0) = v(t_0)$, if $\|\dot{v}(t)\| \leq M_v$ with constant $M_v > 0$, then filter error $\tilde{v}(t) = \bar{v}(t) - v(t)$ satisfies $\|\tilde{v}(t)\| \leq \frac{M_v}{\lambda_{\min}(\tau)}$ for $t \geq t_0$.

Lemma 1 shows that filter error can be limited to an arbitrarily small area by selecting sufficiently large $\lambda_{\min}(\tau)$. The following theorem shows the coordination performance of safety and stability for the closed-loop control system with multiple safety constraints with different relative degrees.

Theorem 2: Let us consider high-order strict-feedback nonlinear affine control system (16) with predesigned desired trajectory y_d and multiple safety constraints with different relative degrees: $h_k(\bar{x}_{r_k}) \geq 0$, $k = 1, \dots, m$. Under Assumptions 1–3, NIVC, MIVC, and SIVC are obtained from (28)–(30), respectively. The direct control input is (33). If system states $\bar{x}_n(t) \in \Omega_x$ for $t \geq t_0$ with Ω_x being a sufficiently large compact set, then initial conditions satisfy $(\bar{x}_r(t_0), z_B(t_0)) \in \bigcap_{k=1}^m (\bigcap_{j=1}^{r-r_k} \mathcal{C}_{j,k} \cap \mathcal{C}_{E,k})$, and parameters satisfy

$$\rho_2 \triangleq \min_{r+1 \leq i \leq n} \lambda_{\min}(\kappa_i) > \frac{l_\gamma}{2} \quad (35)$$

where $l_\gamma = \max_{1 \leq k \leq m} l_{\gamma,k}$. Then, EMB-CBF-SCC achieves the coordination of safety and stability, i.e., forward invariance of a set which is slightly larger than $\bigcap_{k=1}^m (\bigcap_{j=1}^{r-r_k} \mathcal{C}_{j,k} \cap \mathcal{C}_{E,k})$ can be ensured and the tracking error of predesigned desired trajectory z_1 converges to a small neighborhood of the origin for the system without safety constraints.

Proof: First, we prove the bounded property of all filter errors. $\bar{x}_n(t) \in \Omega_x$ means that $\bar{x}_i$ ($i = 1, \dots, n$) is bounded.

For TLS, we can use mathematical induction to prove that the following conditions are true for $i = 1, \dots, r - 1$: 1) v_i is bounded; 2) $\dot{v}_i$ is bounded with $M_i = \max_{\bar{x}_n(t) \in \Omega_x} \|\dot{v}_i\|$;

3) $\tilde{v}_i$ is bounded with $\|\tilde{v}_i\| \leq \frac{M_i}{\lambda_{\min}(\tau_i)}$. The proof process can refer to [18], which uses Lemma 1. Further, if the solution of EMB-CBF-QP (29) is $\beta_r = v_r$, then similarly we can conclude that β_r and $\tilde{\beta}_r$ are all bounded. Otherwise, if $\beta_r \neq v_r$, then we can approximately assume that β_r and $\tilde{\beta}_r$ are all bounded in practice, thus $\|\beta_r\| \leq \frac{M_r}{\lambda_{\min}(\tau_r)}$ and $M_r = \max_{\bar{x}_n(t) \in \Omega_x} \|\beta_r\|$. For BLS, we can again use mathematical induction to prove that the conditions above are true for $i = r + 1, \dots, n - 1$.

Therefore, all filter errors are bounded and can be kept in an arbitrarily small neighborhood of the origin by selecting sufficiently large diagonal elements of τ_i .

Next, according to Definition 6, the proof of the coordination of safety and stability involves the following two aspects.

1) *Stability of the system without safety constraints:* Substituting (26)–(28) into TLES (19) and taking the time derivative of $V_T(z_T)$, one has

$$\begin{aligned} \dot{V}_T = & - \sum_{i=1}^r z_i^\top \kappa_i z_i + z_r^\top g_r z_{r+1} - \delta z_r^\top g_r g_r^\top z_r \\ & + \sum_{i=1}^{r-1} z_i^\top g_i \tilde{v}_i + z_r^\top g_r \tilde{\beta}_r + z_r^\top g_r (\beta_r - v_r). \end{aligned} \quad (36)$$

Substituting (31)–(33) into BLES (20) and taking the time derivative of $V_B(z_B)$, one has

$$\dot{V}_B = - \sum_{i=r+1}^n z_i^\top \kappa_i z_i - \frac{1}{4\delta} z_{r+1}^\top z_{r+1} + \sum_{i=r+1}^{n-1} z_i^\top g_i \tilde{v}_i. \quad (37)$$

The Lyapunov function for the entire system is chosen as

$$V(z_T, z_B) = V_T(z_T) + V_B(z_B). \quad (38)$$

For the system without safety constraints, the solution of EMB-CBF-QP (29) is $\beta_r = v_r$. According to (36) and (37)

$$\begin{aligned} \dot{V} = & - \sum_{i=1}^n z_i^\top \kappa_i z_i + z_r^\top g_r z_{r+1} - \delta z_r^\top g_r g_r^\top z_r - \frac{1}{4\delta} z_{r+1}^\top z_{r+1} \\ & + \sum_{i=1}^{r-1} z_i^\top g_i \tilde{v}_i + z_r^\top g_r \tilde{\beta}_r + \sum_{i=r+1}^{n-1} z_i^\top g_i \tilde{v}_{r+1} \\ \leq & -\rho_1 \sum_{i=1}^n z_i^\top z_i + \zeta_1 \sum_{i=1}^{n-1} z_i^\top z_i + \frac{1}{4\zeta_1} \sum_{i=1}^{n-1} \varepsilon_i^2 \\ \leq & -a_1 V + b_1 \end{aligned} \quad (39)$$

where $\rho_1 = \min_{1 \leq i \leq n} \lambda_{\min}(\kappa_i)$ and constant ζ_1 satisfies $0 < \zeta_1 < \rho_1$. According to the bounded property of filter errors, we can select $\|g_i \tilde{v}_i\| \leq \varepsilon_i$ ($i = 1, \dots, r - 1, r + 1, \dots, n - 1$) and $\|g_r \tilde{\beta}_r\| \leq \varepsilon_r$, $a_1 = 2(\rho_1 - \zeta_1)$, $b_1 = \frac{1}{4\zeta_1} \sum_{i=1}^{n-1} \varepsilon_i^2$. Further, it can be obtained that $V(t) \leq (V(0) - \frac{b_1}{a_1})e^{-a_1(t-t_0)} + \frac{b_1}{a_1}$, thus $\|y - y_d\| \leq \sqrt{\frac{2b_1}{a_1}}$ as $t \rightarrow \infty$, which means tracking error z_1 converges to a small neighborhood of the origin, and error bound decreases as ρ_1 increases.

In particular, the system without safety constraint is exponentially stable if filter errors can be ignored, i.e., $\varepsilon_i = 0$, then $b_1 = 0$, $a_1 = 2\rho_1$, $\dot{V} \leq -2\rho_1 V$, $y_1 \rightarrow y_d$ as $t \rightarrow \infty$.

2) *Safety of the system with safety constraints:* We discuss the situation that there always exists feasible solution of EMB-CBF-QP (29) during the entire time period. From the globally Lipschitz property of $\gamma_{r-r_k+1,k}$ one has

$$\gamma_{r-r_k+1,k}(\psi_{r-r_k,k} - \mu V_B) - \gamma_{r-r_k+1,k}(\psi_{r-r_k,k}) \geq -\mu l_{\gamma,k} V_B. \quad (40)$$

According to (23), (29), (37), (40), and Cauchy's inequality

$$\begin{aligned} & \dot{H}_k(\bar{x}_r, z_B, \bar{\beta}_r, \dot{\beta}_r, u) + \gamma_{r-r_k+1,k}(H_k(\bar{x}_r, z_B)) \\ & \geq \frac{\partial \psi_{r-r_k,k}}{\partial x_r} g_r(\bar{\beta}_r + z_{r+1}) + \frac{2\delta}{\mu} \frac{\partial \psi_{r-r_k,k}}{\partial x_r} g_r g_r^\top \left(\frac{\partial \psi_{r-r_k,k}}{\partial x_r} \right)^\top \\ & \quad + \gamma_{r-r_k+1,k}(\psi_{r-r_k,k} - \mu V_B) - \gamma_{r-r_k+1,k}(\psi_{r-r_k,k}) \\ & \quad + \frac{\mu}{4\delta} z_{r+1}^\top z_{r+1} + \mu \sum_{i=r+1}^n z_i^\top \kappa_i z_i - \mu \sum_{i=r+1}^{n-1} z_i^\top g_i \tilde{v}_i \\ & \geq -\mu l_{\gamma,k} V_B + \mu \rho_2 \sum_{i=r+1}^n z_i^\top z_i - \mu \zeta_2 \sum_{i=r+1}^{n-1} z_i^\top z_i - \mu b_2 \\ & \geq \mu(2\rho_2 - 2\zeta_2 - l_{\gamma}) V_B - \mu b_2 \geq -\mu b_2 \end{aligned} \quad (41)$$

where $b_2 = \frac{M_r^2}{4\delta\lambda_{\min}^2(\tau_r)} + \frac{1}{4\zeta_2} \sum_{i=r+1}^{n-1} \varepsilon_i^2$ and constant ζ_2 satisfies $0 < \zeta_2 < \rho_2 - \frac{l_{\gamma}}{2}$. We define new functions

$$\begin{aligned} \Phi_{j,k} &= \psi_{j-1,k} - \gamma_{j,k}^{-1} \circ \gamma_{j+1,k}^{-1} \circ \dots \circ \gamma_{r-r_k+1,k}^{-1}(-\mu b_2) \\ j &= 1, \dots, r - r_k + 1 \\ \Phi_k &= H_k - \gamma_{r-r_k+1,k}^{-1}(-\mu b_2) \end{aligned} \quad (42)$$

and their superlevel sets are defined as follows:

$$\begin{aligned} \mathcal{D}_{j,k} &= \{\bar{x}_r \in \mathcal{W}_r, z_B \in \mathbb{R}^{|\mathcal{X}_B|} : \Phi_{j,k}(\bar{x}_r, z_B) \geq 0\} \\ j &= 1, \dots, r - r_k + 1 \\ \mathcal{D}_{E,k} &= \{\bar{x}_r \in \mathcal{W}_r, z_B \in \mathbb{R}^{|\mathcal{X}_B|} : \Phi_k(\bar{x}_r, z_B) \geq 0\} \end{aligned} \quad (43)$$

where $\mathcal{D}_{j,k}$ is slightly larger than $\mathcal{C}_{j,k}$ and $\mathcal{D}_{E,k}$ is slightly larger than $\mathcal{C}_{E,k}$.

According to (41), when $\Phi_k < 0$, one has $\dot{\Phi}_k = \dot{H}_k \geq -\gamma_{r-r_k+1,k}(H_k) - \mu b_2 > 0$, and when $\Phi_k = 0$ one has $\dot{\Phi}_k \geq 0$. According to Nagumo's theorem [5] the forward invariance of $\mathcal{D}_{E,k}$ can be ensured. Since $(\bar{x}_r(t_0), z_B(t_0)) \in \mathcal{C}_{E,k} \subset \mathcal{D}_{E,k}$, then $\Phi_{r-r_k+1,k} = \psi_{r-r_k,k} - \gamma_{r-r_k+1,k}^{-1}(-\mu b_2) \geq \mu V_B \geq 0$ as $t \geq t_0$, so $\mathcal{D}_{r-r_k+1,k}$ is forward invariant. We next recursively prove the forward invariance of $\mathcal{D}_{j,k}$. If set $\mathcal{D}_{j+1,k}$ is forward invariant, then for $\mathcal{D}_{j,k}$, and one can obtain from (21) and (42) that when $\Phi_{j,k} < 0$, then $\dot{\Phi}_{j,k} = \psi_{j,k} - \gamma_{j,k}(\psi_{j-1,k}) > \psi_{j,k} - \gamma_{j+1,k}^{-1} \circ \dots \circ \gamma_{r-r_k+1,k}^{-1}(-\mu b_2) = \Phi_{j+1,k} \geq 0$. When $\Phi_{j,k} = 0$, then $\dot{\Phi}_{j,k} = \dot{\Phi}_{j+1,k} \geq 0$, so set $\mathcal{D}_{j,k}$ is also forward invariant, since $(\bar{x}_r(t_0), z_B(t_0)) \in \mathcal{C}_{j,k} \subset \mathcal{D}_{j,k}$. Overall, EMB-CBF-SCC ensures forward invariance of set $\bigcap_{k=1}^m (\bigcap_{j=1}^{r-r_k} \mathcal{D}_{j,k} \cap \mathcal{D}_{E,k})$, which is slightly larger than $\bigcap_{k=1}^m (\bigcap_{j=1}^{r-r_k} \mathcal{C}_{j,k} \cap \mathcal{C}_{E,k})$.

In particular, if filter errors can be ignored, i.e., $b_2 = 0$, $\dot{H}_k + \gamma_{r-r_k+1,k}(H_k) \geq 0$, according to [6, Thm. 2], the forward invariance of set $\mathcal{C}_{E,k}$ can be ensured, thus $\psi_{r-r_k,k}(\bar{x}_r) \geq \mu V_B(z_B) \geq 0$ as $t \geq t_0$, the forward invariance of $\bigcap_{j=1}^{r-r_k} \mathcal{C}_{j,k}$ can be obtained from [9, Thm. 1], so

TABLE II
SAFETY AND STABILITY ANALYSIS FOR SIMULATION

		Ours	HOCBF	SBP	SVT
Safety (MSCV)	Case 1:[00.00, 20.00]s	2.4e-3	2.9e-3	-6.1e-3	-8.9e-4
	Case 2:[02.24, 10.00]s	3.6e-2	8.0e-4	1.4e-2	6.6e-4
	Case 2:[10.00, 20.00]s	3.7e-3	-5.4e-2	-1.3e-2	-4.4e-2
Stability (RMSE)	Case 1:[05.73, 06.84]s	3.4e-3	4.2e-3	1.8e-2	1.8e-2
	Case 2:[05.73, 06.84]s	4.1e-3	4.2e-3	1.6e-2	1.7e-2
	Case 2:[15.15, 16.26]s	4.8e-3	1.3e-2	5.5e-2	1.3e-2

* MSCV: minimum safety constraint value (angle constraint).
RMSE: root-mean-square-error.

set $\bigcap_{k=1}^m (\bigcap_{j=1}^{r-r_k} \mathcal{C}_{j,k} \cap \mathcal{C}_{E,k})$ is forward invariant. Initial state starting from outside eventually returns to this set, thus the system with safety constraints is safe.

EMB-CBF-SCC achieves coordination of safety and stability. Proof of Theorem 2 is complete.

Remark 4: EMB-CBF-QP (29) may have no feasible solution under multiple constraints, which is rarely discussed in existing works. A practical solution to ensure feasibility of QP in these cases is to introduce relaxation variables. The detailed strategy we propose is to transform (29) into the following form:

$$\begin{aligned} \beta_r(\bar{x}_r) &= \arg \min_{\beta_r \in \mathcal{X}_r, [\vartheta_1, \dots, \vartheta_m] \in \mathbb{R}^m} \|\beta_r - v_r\|^2 + \sum_{k=1}^m p_k^2 \vartheta_k^2 \\ \text{s.t. } & \psi_{r-r_k,k}(\bar{x}_r, \beta_r, 0) + \gamma_{r-r_k+1,k}(\psi_{r-r_k,k}(\bar{x}_r)) + \vartheta_k \geq \\ & \frac{2\delta}{\mu} \frac{\partial \psi_{r-r_k,k}}{\partial x_r} g_r g_r^\top \left(\frac{\partial \psi_{r-r_k,k}}{\partial x_r} \right)^\top, k = 1, \dots, m \end{aligned} \quad (44)$$

where ϑ_k is the relaxation variable of the k th constraint and $p_k > 0$ is the corresponding penalty factor. Equation (44) ensures solvability of EMB-CBF-QP. A detailed example is also presented in Case 2 of numerical simulation.

V. APPLICATION TO ROBOTIC SYSTEMS

A. Numerical Simulation of a Single-Link Manipulator

We consider a direct-driven single-link manipulator [19] with third-order strict-feedback nonlinear affine form

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \frac{S}{P} \sin x_1 - \frac{S}{P} x_2 + \frac{0.1}{P} x_1 \sin x_2 + \frac{1}{P} x_3 \\ \dot{x}_3 = -\frac{K_B}{L} x_2 - \frac{R}{L} x_3 + \frac{1}{L}(1 + x_1^2)u \end{cases} \quad (45)$$

where x_1 , x_2 , and x_3 are angle, angular velocity, and armature current, respectively. System parameters are the same as in [19]. The predesigned desired trajectory is $y_d = \sin t$. Maximum angular acceleration $\alpha_{\max} = 4$ is a natural constraint.

1) *Case 1: Single Constraint With Safe Initial State:* To limit angle to the area of $|x_1| \leq x_{1\max}$ with $x_{1\max} = \pi/4$, we select a sufficient but unnecessary condition with relative degree two: $h(x_1, x_2) = x_{1\max} - \text{sign}(x_2)x_1 - 0.5x_2^2/\alpha_{\max}$. The system starts from safe initial state $[x_1(0), x_2(0), x_3(0)]^\top = [\pi/8, 0, 0]^\top$, controller parameters are $\kappa_1 = 5$, $\kappa_2 = 8$, and $\kappa_3 = 10$. Filter parameters are $\tau_1 = \tau_2 = \tau_3 = 80$. EMB-CBF-QP parameters are $\delta = 1/32$, $\mu = 0.1$, and $\gamma_{1,1}(a) = 3a$.

Simulation results are shown in Fig. 3 and Table II. EMB-CBF-SCC tracks the desired trajectory under the premise of meeting the safety constraint. PPC [20] is invalid because the

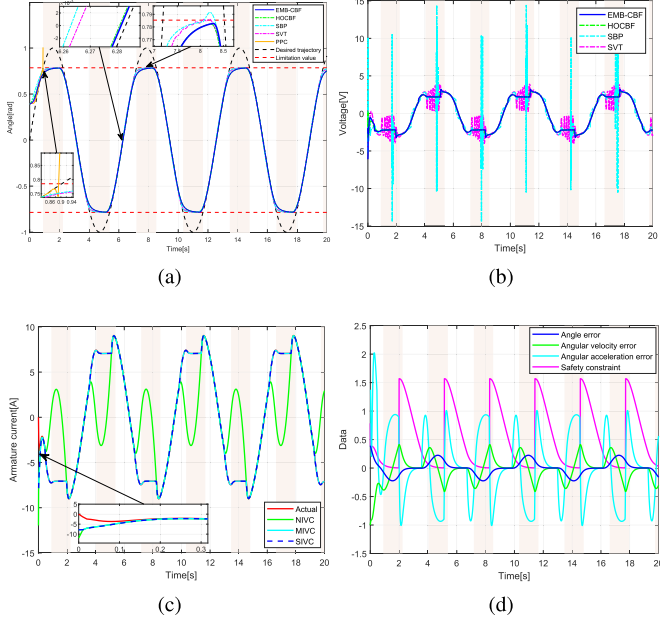


Fig. 3. Case 1: The desired trajectory violates angle limitation in the pale red area. (a) Joint angle curves of the different algorithms. (b) Control input curves of the different algorithms. (c) Safety modification process (NIVC $\Rightarrow$ MIVC $\Rightarrow$ SIVC) of EMB-CBF-SCC to armature current. (d) State errors and safety constraint value curves of EMB-CBF-SCC.

nonlinear transformation function brings large fluctuation near the safety boundary, and causes opposite movement outside it. Compared with other CBF-based methods, EMB-CBF-SCC achieves similar performance as HOCBF [10], but involves less computation. In terms of safety, EMB-CBF-SCC keeps farther safe distance from boundary than SBP [14] and SVT [15]. SBP and SVT slightly overtake the boundary, and their control input curves fluctuate greatly in the conflict areas of safety and stability. In terms of stability, our method exhibits less tracking error in areas far away from the safety boundary.

2) Case 2. Multiple Constraints With Unsafe Initial State: The manipulator should satisfy angle, angular velocity, and angular acceleration constraints simultaneously. Safety constraints are

$$h_1(x_1) = \begin{cases} x_{1\max}^2 - x_1^2 & \text{if } |x_1| > x_{1\text{act}} \\ x_{1\max}^2 - x_{1\text{act}}^2 & \text{other} \end{cases}$$

and $h_2(x_2) = x_{2\max}^2 - x_2^2$, with $x_{1\text{act}} = \pi/6$ and $x_{2\max} = 2$. They have different relative degrees 3 and 2. The form of h_1 further reduces computational burden, because EMB-CBF-QP does not need to be solved in the inactive area. The system starts from unsafe initial state $[x_1(0), x_2(0), x_3(0)]^T = [\pi/2, 0, 0]^T$, and suffers from random noise and additional disturbance after 10 s. The form of EMB-CBF-QP with relaxation variables is

$$\begin{aligned} \beta_2(x_1, x_2) = \arg \min_{(\beta_2, \vartheta_1, \vartheta_2) \in \mathbb{R}^3} & (\beta_2 - \vartheta_2)^2 + p_1^2 \vartheta_1^2 + p_2^2 \vartheta_2^2 \\ \text{s.t. } \frac{\partial h_1}{\partial x_1} (f_2 + g_2 \beta_2) + \frac{\partial^2 h_1}{\partial x_1^2} x_2^2 + (\gamma_{1,1} + \gamma_{2,1}) \frac{\partial h_1}{\partial x_1} x_2 \\ & + \gamma_{2,1} \gamma_{1,1} h_1 + \vartheta_1 \geq \frac{2\delta}{\mu} g_2^2 \left(\frac{\partial h_1}{\partial x_1} \right)^2 \end{aligned}$$

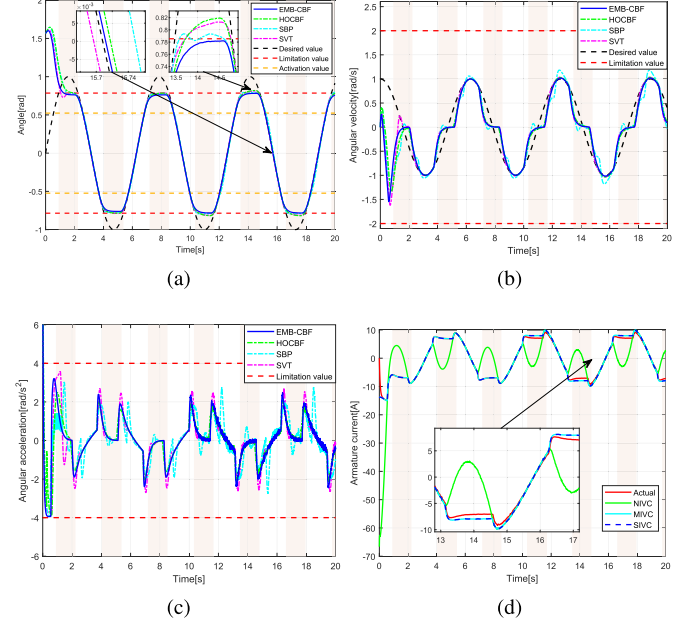


Fig. 4. Case 2: The desired trajectory violates angle limitation in the pale red area. (a) Joint angle curves of the different algorithms. (b) Joint angular velocity curves of the different algorithms. (c) Joint angular acceleration curves of the different algorithms. (d) Safety modification process (NIVC $\Rightarrow$ MIVC $\Rightarrow$ SIVC) of EMB-CBF-SCC to armature current.

$$\begin{aligned} \frac{\partial h_2}{\partial x_2} (f_2 + g_2 \beta_2) + \gamma_{1,2} h_2 + \vartheta_2 & \geq \frac{2\delta}{\mu} g_2^2 \left(\frac{\partial h_2}{\partial x_2} \right)^2 \\ \frac{-\alpha_{\max} - f_2}{g_2} \leq \beta_2 \leq \frac{\alpha_{\max} - f_2}{g_2} \end{aligned} \quad (46)$$

where $f_2 = \frac{E}{P} \sin x_1 - \frac{S}{P} x_2 + \frac{0.1}{P} x_1 \sin x_2$ and $g_2 = \frac{1}{P}$, ϑ_1 and ϑ_2 are the relaxation variables, $p_1 = 10^6$ and $p_2 = 10^3$ are the corresponding penalty factors, and all class $\mathcal{K}$ functions are selected as linear forms.

Simulation results are shown in Fig. 4 and Table II. EMB-CBF-SCC quickly returns the system to the angle limitation region even if starting from an unsafe initial state, and then meets all constraints. EMB-CBF-SCC achieves arrival speed of safe set faster than HOCBF, and with smaller change of angular velocity and acceleration than SBP and SVT within the safe set. Compared with other CBF-based methods, ours keeps the maximum safe distance from the safety boundary, and is the only algorithm that does not violate angle constraint when the system suffers from noise and disturbance. It also has the highest control accuracy in the region of inactive angle constraint. Therefore, EMB-CBF-SCC achieves better coordination performance of safety and stability, and has stronger robustness than these state-of-the-art methods.

B. Physical Experiment: Micro Mobile Robot

We apply EMB-CBF-SCC to our self-developed micro mobile robot (ICSBOT) for avoiding static and dynamic unknown obstacles. Hardware setup is shown in Fig. 5. The robot dynamics

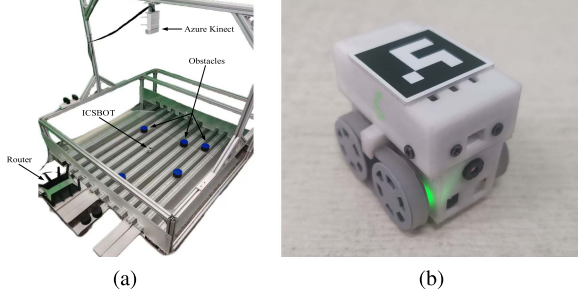


Fig. 5. Hardware setup. (a) Experimental platform ($1.3 \times 1.4 \times 1.5$ m) with movable obstacles. (b) ICSBOT ($3.8 \times 5.0 \times 4.5$ cm).

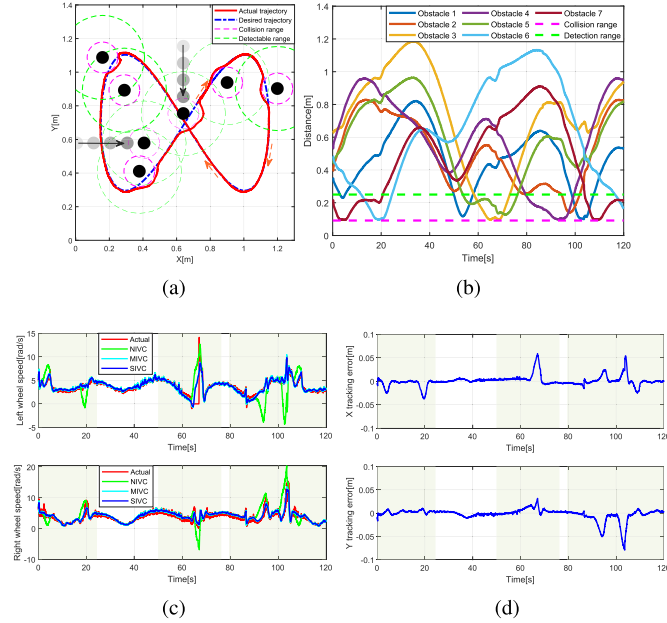


Fig. 6. Experimental results after 2.10 s. (a) Robot head trajectory. (b) Distance between robot head and obstacles (obstacles 1 and 2 are dynamic, all others static). (c) Safety modification process to wheel speed (the pale green area indicates the presence of obstacles within the robot's detection range). (d) Tracking error of predesigned desired trajectory.

model has the following second-order form:

$$\begin{cases} \dot{p}_h = r_w \begin{bmatrix} \cos \theta & -l_h \sin \theta \\ \sin \theta & l_h \cos \theta \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{l_b} & \frac{1}{l_b} \end{bmatrix} w \\ \dot{w} = -\frac{1}{r_w} \left(\frac{c_m c_e \eta_m^2}{R_m} + b_w \right) \Lambda w + \frac{c_m \eta_m}{R_m r_w} \Lambda u \end{cases} \quad (47)$$

where p_h is the position of robot head, θ is the heading angle, w is the wheel speed vector, and u is the input voltage vector. l_h , l_b , r_w , c_e , c_m , R_m , η_m , b_w , and Λ are all constants associated with ICSBOT. The predesigned desired trajectory for robot center is $p_{cd} = [0.65 + 0.5 \sin(0.06t), 0.7 + 0.4 \sin(0.12t)]^T$. The collision avoidance constraint for k th obstacle is

$$h_k(p_h) = \begin{cases} \|p_h - p_{o,k}\|^2 - (R_h + R_o)^2 & \text{if } k \in \mathcal{S} \\ R_s^2 - (R_h + R_o)^2 & \text{if } k \notin \mathcal{S} \end{cases}$$

with collision radius of obstacles $R_o = 0.038$ m, collision radius of robot head $R_h = 0.0535$ m, and detection radius of robot

head $R_s = 0.25$ m. $p_{o,k}$ is the position of k th obstacle and $\mathcal{S} = \{k : \|p_h - p_{o,k}\|^2 \leq R_s^2\}$ the obstacles set within robot's detection range. The maximum wheel speed is 15.7 rad/s.

Experimental results are shown in Fig. 6. The robot completely avoids collision with all the static and dynamic unknown obstacles within maximum wheel speed constraint, and quickly converges to the desired trajectory in the area away from the obstacles. Thus, the SCC goal is achieved. HOCBF fails because it has no feasible solution for a long time. The safety modification of EMB-CBF-QP is minimal. When it is inactive, $t \in [25, 49]$ s, RMSE of tracking error of the desired trajectory is only 0.0053 m. Therefore, the great effectiveness of EMB-CBF-SCC is clearly verified.

Remark 5: Our algorithm is fully deployed in the ESP32-CAM chip of ICSBOT, with control frequency 30 Hz. This means that EMB-CBF-SCC has simple computation, and is perfectly suitable for embedded platforms.

VI. CONCLUSION

This article introduces EMB-CBF and applies it to the SCC of high-order strict-feedback nonlinear affine systems with multiple different high-relative-degree safety constraints. Compared with existing CBF-based methods, it significantly reduces differential order and computational burden by using command filters, as well as it exhibits improved performance. We theoretically prove coordination performance of safety and stability considering filter errors. The advantages of EMB-CBF-SCC are clearly demonstrated in both simulations and real experiments with two types of robots.

Future research will focus on extending the proposed framework to SCC of multiagent systems with complex perturbations.

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